

Practice Problems Sheet IX

(Solving Recurrence Relations)

Q1 Prove the identity relating the Fibonacci numbers and binomial coefficients:

$$f_{n+1} = \binom{n}{0} + \binom{n-1}{1} + \dots + \binom{n-k}{k}, \text{ for all integers } n \geq 1 \text{ and } k = \lfloor n/2 \rfloor$$

[Hint: Define $a_n = \binom{n}{0} + \binom{n-1}{1} + \dots + \binom{n-k}{k}$, for $n \geq 0$.

Then show that sequences $(a_n)_{n \in \mathbb{N}}$ and $(f_n)_{n \in \mathbb{N}}$ satisfy the same recurrence and have the same initial conditions.]

Q2 Find all solutions of recurrence relation

$$a_n = 4a_{n-1} - 4a_{n-2} + (n+1)2^n$$

Q3 Solve the simultaneous recurrence relations

$$a_n = 3a_{n-1} + 2b_{n-1}$$

$$b_n = a_{n-1} + 2b_{n-1}$$

with $a_0 = 1$ and $b_0 = 2$.

Q4 In class we found explicit formula for n^{th} Fibonacci number f_n as

$$f_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^n.$$

Using this show that f_n is the nearest integer to $\frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^n$.

Q5 Use substitution method to solve the recurrence relation $T(n) = n[T(n/2)]^2$, when n is a power of 2, with initial condition $T(1) = 6$.

End
